Amogh Gulati

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Education

NYU Courant Institute of Mathematical Sciences — New York, NY | Sep 2025 – May 2027
M.S. Computer Science — GPA: 3.8/4.0

IIIT Delhi — New Delhi, India | Aug 2017 – May 2021
B.Tech. Computer Science and Engineering — GPA: 8.71/10

Skills

Languages: Python, C++, SQL, Bash
ML & Data: PyTorch (DDP), ONNX, Weights & Biases, Xarray, Zarr, Pandas, NumPy
Infrastructure: Slurm, Docker, AWS (Lambda, SageMaker, S3, SQS, CloudFront), Elasticsearch, Kibana

Experience

M^{2}LInES, NYU Courant | *Graduate Research Assistant* — New York, NY | Oct 2025 – Present

- Improved Samudra, an ML ocean emulator: skip-connection dropout reduced rollout RMSE by up to **15%** on velocity and **8.5%** on sea surface height; ConvNeXt–axial attention blocks reduced RMSE across all variables and depths.
- Developing replay-buffer training and spectral losses to address spatial smoothing and long-term trend errors in 100-year ocean rollouts.

Gan.ai (Sequoia-backed) | *Founding ML Engineer* — Noida, India | Jan 2021 – Oct 2024

- Built and owned the company's ML stack from scratch, spanning TTS model training, production inference, and deployment; led the ML infrastructure team.
- Trained and deployed TTS models (NaturalSpeech2, FastSpeech2, Tortoise) for English and 6 Indic languages, serving **200k+ daily inferences**.
- Designed AWS deployment pipelines using Lambda, SageMaker, Docker, and ONNX, reducing cold-start latency from **60s to 4s**.
- Owned delivery of personalized video ads for Samsung and AI-generated birthday songs for Cadbury.
- Established security and operational processes and compiled audit evidence for the company's **SOC 2 Type II** compliance.
- Co-inventor on **2 pending US patent applications** for AI voice-over and multimedia generation.

Precog Research Group, IIIT Delhi | *Undergraduate Researcher* — New Delhi, India | Jun 2019 – Jan 2021

- Built ELK pipelines to automate collection of **100M+ tweets** for social media research.
- Analyzed emotions and misinformation in COVID-19 and political discussions on social media, resulting in publication (below).

Qualcomm | *Software Engineering Intern* — Remote | May 2020 – Aug 2020

- Extended debugging features in the Hexagon IDE and automated build-goal validation with Python.

Publications & Patents

Google Scholar

- Varadhan, P. S., **Gulati, A.**, Sankar, A., Anand, S., et al. (2025). Rethinking MUSHRA: Addressing Modern Challenges in Text-to-Speech Evaluation. *TMLR*.
- Aggrawal, P., Jolly, B. L. K., **Gulati, A.**, et al. (2021). Psychometric Analysis and Coupling of Emotions Between State Bulletins and Twitter in India During COVID-19 Infodemic. *Frontiers in Communication*.
- Prabhu, A., Guhathakurta, D., Subramanian, M., Reddy, M., Sehgal, S., Karandikar, T., **Gulati, A.**, et al. (2021). Capitol (Pat)riots: A Comparative Study of Twitter and Parler. *arXiv:2101.06914*.
- **Gulati, A.**, Joshi, B., Jain, C., and Shukla, J. (2020). It's Not What They Play, It's What You Hear: Understanding Perceived vs. Induced Emotions in Hindustani Classical Music. *ICMI '20 Companion*.
- **Gulati, A.**, Siddhartha, S., Barman, M. P., Bhatia, A., and Bhooshan, S. (2025). Voiced-over Multimedia Track Generation. *US Patent App. 18/842,687* (pending).
- Siddhartha, S., Bhatia, A., **Gulati, A.**, Barman, M. P., and Bhooshan, S. (2025). Seamless Multimedia Integration. *US Patent App. 19/242,615* (pending).